



Sanjivani Rural Education Society's
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(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 3 – Polymorphism and Virtual Function
Topic – 3.1 Pure Virtual Function and Abstract Class



Pure Virtual Function

- Pure virtual functions are used if a function doesn't have any use in the base class but the function must be implemented by all its derived classes.

- Let's take an example,

Suppose, we have derived Triangle, Square and Circle classes from the Shape class, and we want to calculate the area of all these shapes.

- In this case, we can create a pure virtual function named `calculateArea()` in the Shape base class. Since it's a pure virtual function, all derived classes Triangle, Square and Circle must include the `calculateArea()` function with their implementations.
- A pure virtual function doesn't have the function body and it must end with `= 0`.

- **Example using pure virtual function:**

```
class Shape
```

```
{
```

```
    public:
```

```
        virtual void calculateArea() = 0;
```

```
};
```

Note: The = 0 syntax doesn't mean we are assigning 0 to the function. It's just the way we define pure virtual function.



Abstract Class

- A class that contains a pure virtual function is known as an abstract class.
- In the previous example, the class Shape is an abstract class.
- We cannot create objects of an abstract class.
- However, we can derive classes from abstract class, and use their data members and member functions (except pure virtual functions).



Example: program to calculate the area of a square and a circle using Abstract Class and Pure Virtual Function

```
// Abstract class
class Shape
{
    protected:
        float dimension;
    public:
        void getDimension()
        {
            cin >> dimension;
        }
        // pure virtual Function
        virtual float calculateArea()=0;
};
class Square : public Shape
{
    public:
```

```
float calculateArea()
{
    return dimension * dimension;
};
class Circle : public Shape
{
    public:
        float calculateArea()
        {
            return
            3.14*dimension*dimension;
        }
    void main()
    {
        Square s;
```

```
Circle c;
cout<<"Enter length of a square: ";
s.getDimension();
cout<<"Area of square:";
s.calculateArea() ;
cout<<"Enter radius of a circle:";
c.getDimension();
cout<<"Area of circle: ";
c.calculateArea() ;
return 0;
}
```

Output:

```
Enter length of a square: 4
Area of square: 16
Enter radius of a circle: 5
Area of circle: 78.5
```

- In this program, `virtual float calculateArea() = 0;` inside the Shape class is a pure virtual function.
- That's why we must provide the implementation of `calculateArea()` in both of our derived classes, or else we will get an error.

Thank You..